

Replication Report: Szmolka *et al.* (2023) — *Emergence and Genomic Features of a mcr-1 Escherichia coli* *from Duck in Hungary*

Ollie (OpenClaw AI) — BVBRC-104 Replication
Free-endpoint agent pass; Argo `argo:claude-opus-4.7` + Argo `argo:gpt-5.2` LLM judge

Original run: 2026-07-05 (US-Central)
LaTeX backfill: 2026-07-05 (RESUME-mode 8-artifact backfill)

Abstract

This is the LaTeX-detailed replication report for Szmolka *et al.* (2023, *Antibiotics* 12(10):1519, DOI [10.3390/antibiotics12101519](https://doi.org/10.3390/antibiotics12101519), PMID 37887221, PMC10604428). We independently pulled all six deposited NCBI Nuccore records (CP134085–CP134090) for strain Ec45-2020, re-ran the paper’s genomic in-silico pipeline (mlst 2.33.1 Warwick scheme, AMRFinderPlus 4.2.7, abricate + PlasmidFinder + ecoh, BLAST+ 2.17.0 vs a fresh 17-plasmid IncX4/mcr-1 reference set), and validated the paper’s central claims. **Verdict: REPLICATED** (LLM judge `argo:gpt-5.2`: verdict=REPLICATED, coverage=85%, agreement=100%, confidence=high). Coverage <100% because the paper’s cgMLST tree across ~500 BV-BRC strains (C10) and the wet-lab conjugation experiment (C11) were declared out of scope. This LaTeX report additionally provides an honest critique of the replication’s evidence strength (§5) and five heavy-duty open questions grounded in re-reading the paper (§7).

1 Paper Summary

Szmolka *et al.* report the **first mcr-1-positive E. coli isolate from a duck in Hungary**, strain *Ec45-2020*, recovered in 2020 from a 6-day-old duckling that died from a suspected systemic infection on a Hungarian farm. Systematic PCR screening of 479 *E. coli* isolates from 483 poultry cloacal/caecal samples across 34 farms and 4 slaughterhouses yielded exactly *one* mcr-1-positive strain, confirmed by Sanger sequencing.

The strain was whole-genome-sequenced in parallel on Illumina MiSeq (2×250 bp) and ONT MinION Mk1C, with hybrid assembly via Unicycler 0.5.0 + Nanopolish 0.14.0, producing a complete circular genome deposited under BioProject PRJNA1012593. Key genomic findings:

- **Serotype** O55:H10; **MLST** ST162 (Warwick scheme).
- **Phenotype** Amp-Chl-Cip-Col-Sul-Tet-Tmp (multi-drug resistant); **colistin MIC** 8 µg/mL.
- **Genome layout**: chromosome CP134085 (~4.97 Mb, paper wrote 4,966,963 bp; the NCBI record is 4,967,063 bp — a 100 bp typo) + 5 circular plasmids CP134086–CP134090.
- **mcr-1.1** sits alone on the 33,541 bp *IncX4* plasmid pEc45-2020-33kb (CP134089).
- A 254 kb *IncH* MDR plasmid (CP134088) carries *dfrA12*, *aadA1/2*, *sul3*, *cmlA1*, *floR*, *qnrS1*.

- A 190 kb *IncF* hybrid plasmid (CP134087) carries virulence genes (hly, tra, iut, iuc) and additional AMR including the duplicated blaTEM-135--sul2--tet(A), (M) cluster (also on CP134088).
- The remaining 101 kb + 5 kb plasmids carry no AMR/VF genes (per paper).
- BLASTn of pEc45-2020-33kb vs 26 published IncX4/mcr-1 plasmids: 100% query coverage, 93–98% pairwise identity — “highly conserved backbone” interpretation.
- cgMLST across 238 BV-BRC mcr-1 E. coli genomes clusters Ec45-2020 with Chinese human ST162 strains but not with a Chinese duck strain.
- Wet-lab conjugation: pEc45-2020-33kb is *non-transferable* under the experimental conditions — unexpected because most IncX4/mcr-1 plasmids are conjugative; oriT is not detectable in the sequence.

2 Claims Table

#	Claim	Type	Testable?	Tested?
C1	Chromosome + 5 plasmids deposited under PR-JNA1012593 (CP134085–CP134090) with expected sizes	Data avail-ability	Yes	YES
C2	Strain is ST162 (Warwick scheme)	Genomic	Yes	YES
C3	Serotype O55:H10	Genomic	Yes	YES
C4	mcr-1 sits on 33,541 bp IncX4 plasmid pEc45-2020-33kb (CP134089) and is the exclusive AMR gene on it	Genomic	Yes	YES
C5	254 kb plasmid (CP134088) is IncH and carries dfrA12, aadA1/2, sul3, cmlA1/floR, qnrS1	Genomic	Yes	YES
C6	190 kb hybrid plasmid (CP134087) carries virulence + duplicated blaTEM-135-sul2-tet(A/M) cluster	Genomic	Yes	YES
C7	101 kb (CP134086) + 5 kb (CP134090) plasmids carry no AMR/virulence determinants	Genomic (negative)	Yes	YES (†)
C8	Chromosome has APEC-typical astA, fyuA, hlyE, lpfA virulence	Genomic	Yes	PARTIAL
C9	BLASTn of pEc45-2020-33kb vs public IncX4/mcr-1 plasmids: 100% qcov, 93–98% pident	Genomic comparative	Yes	YES
C10	cgMLST of 504 BV-BRC mcr-1 E. coli clusters Ec45-2020 with Chinese human ST162 strains	Comparative genomics	Yes (heavy)	NO (out of scope)
C11	pEc45-2020-33kb is non-transferable under conjugation conditions	Wet-lab	No	NO (not testable in silico)

† C7 is confirmed only against the AMRFinderPlus curated panel; see §5.

3 Method

All steps executed on 2026-07-05 US-Central. Compute: uicgpu (8×A100, 255-core Xeon, 2 TB RAM, `bvbr14` conda env) for pipeline runs; CherryRd for orchestration + PMC fetch + Argo calls. Free endpoints only (Argo localhost:44497 key=`stevens`). Total wall clock end-to-end ~13 min.

1. **Paper fetch:** MDPI direct HTML was blocked by anti-bot (Access Denied); pivoted to NCBI E-utilities `efetch db=pmc id=10604428 rettype=xml` (124 KB JATS XML), parsed with Python 3.14 `xml.etree.ElementTree` to `work/paper_text.md` (26.8 KB).
2. **Strain assembly:** for each of CP134085–CP134090, `efetch db=nucore rettype=fasta` to `work/genomes/`. Concatenated to `Ec45-2020_all.fasta` (5.63 MB).
3. **MLST:** `mlst 2.33.1` (Torsten Seemann), scheme `ecoli_achtman_4` (= Warwick). Also ran the `ecoli` Pasteur scheme as a control (returned ST355 — different scheme, expected).
4. **AMRFinderPlus 4.2.7** with autoupdated DB, `-O Escherichia --plus` on combined assembly. Wall clock 64 s. Full TSV in `report/evidence/amrfinder_full.tsv`. Partitioned by contig.
5. **PlasmidFinder** via `abricate 1.0.1` DB `plasmidfinder` (488 seqs, 2026-Apr-3).
6. **SerotypeFinder** via `abricate`, DB `ecoh` (597 seqs).
7. **IncX4/mcr-1 reference harvest for backbone-conservation BLAST:** NCBI Nucleotide search `"mcr-1" "IncX4" plasmid "complete sequence" 30000:40000[SLEN]` → 17 real IncX4/mcr-1 plasmid refs (30–40 kb). First attempt without size filter had returned a virus and a fungal mRNA due to term ambiguity — fixed by size filter.
8. **BLASTn:** `makeblastdb + blastn -evalue 1e-50 -outfmt "6 sseqid pident length qstart qend evalue"`. Per-subject non-overlapping HSP merge + weighted pident via `work/blast_summary`. Result JSON in `report/evidence/blast_incX4_v2.summary.json`.
9. **LLM judge** (per Wave-brief hard rule): 7.5 KB prompt covering every paper claim + every rerun evidence block. First 4 attempts to `argo:claude-opus-4.7` returned HTTP 502 (transient Argo Vertex issue); fallback to `argo:gpt-5.2` succeeded on the first call.

4 Results vs Paper

4.1 Genome / plasmid sizes

Replicon	Paper (bp)	This rerun (bp)	Δ
CP134085 chromosome	4,966,963	4,967,063	+100 bp (likely paper typo; 0.002% on 5 Mb)
CP134086 (101 kb)	~101,000	101,848	✓
CP134087 (190 kb)	~190,000	190,488	✓
CP134088 (254 kb)	~254,000	254,224	✓
CP134089 (33 kb)	33,541	33,541	exact
CP134090 (5 kb)	~5,000	5,714	✓

4.2 MLST + serotype

Warwick scheme *ST162* independently rederived (adk 9, fumC 65, gyrB 5, icd 1, mdh 9, purA 13, recA 6). Serotype *O55:H10* independently confirmed (wzy-O55 97.75%/100% cov, wzx-O55 98.59%/100% cov, fliC-H10 99.44%/99.92% cov).

4.3 mcr-1 plasmid content (CP134089, IncX4, 33,541 bp)

AMRFinderPlus returns *exactly one* AMR hit on this contig: **mcr-1.1** (WP_049589868.1), 541/541 aa, 100% identity, 100% coverage, ALLELEX. PlasmidFinder types the contig as **IncX4_1** (CP002895), 100% id, 100% cov. Both consistent with C4.

4.4 254 kb IncH MDR plasmid (CP134088)

All 9 paper-claimed AMR genes recovered at $\geq 99.5\%$ identity, 100% coverage: dfrA12, aadA1, aadA2, sul3, cmlA1, floR, qnrS1, blaTEM-135, tet(A). PlasmidFinder refines the paper’s “IncH” label to **IncHI1A + IncHI1B(R27) + IncFIA(HI1)** — refinement, not disagreement. Additional hits our rerun surfaced beyond the paper’s enumeration: *sil* operon (Sila-P) + *pco* operon (PcoA-E) heavy-metal/silver resistance; qacL biocide efflux; tet(M); sul2 (part of the duplicated cluster).

4.5 190 kb hybrid plasmid (CP134087)

Aerobactin operon *iutA*, *iucA*, *iucB*, *iucC*, *iucD* all present; *traT* conjugation; the *blaTEM-135-sul2-tet(A/M)* cluster *exactly* duplicated between CP134087 and CP134088; additional *mchF* (microcin H47 export) that the paper did not enumerate. The *hly* hemolysin gene claimed by the paper is not called by AMRFinderPlus’s VF panel — see §5.

4.6 101 kb + 5 kb plasmids (CP134086, CP134090)

AMRFinderPlus returns 0 hits on both. Consistent with C7 *as stated by the paper*, but see §5 on why this is a weaker confirmation than the paper implies.

4.7 Chromosome virulence

astA + *lpfA* + *lpfA-O113* variant confirmed on CP134085. Paper’s *fyuA* (yersiniabactin siderophore receptor) is not in the AMRFinderPlus VF panel, but the same-operon transporter subunits *ybtP* + *ybtQ* are detected, confirming the yersiniabactin pathway is present. Paper’s *hlyE* is not in the AMRFinderPlus VF panel; we cannot confirm or refute it with this pipeline alone.

Beyond the paper: we additionally detected chromosomal *gyrA_S83L* + *gyrA_D87N* + *parC_S80I* quinolone-resistance point mutations (explaining the paper-reported ciprofloxacin resistance — the paper did not enumerate the ciprofloxacin-resistance genotype), plus *blaEC* intrinsic class C beta-lactamase, and *acrF/mdtM/emrE* efflux transporters, plus *fdeC/espX1/ariR*.

4.8 IncX4 backbone conservation (C9)

BLASTn of pEc45-2020-33kb (query, 33,541 bp) vs 17 published IncX4/mcr-1 plasmids (30–40 kb range) after non-overlapping HSP merge + weighted pident: *median* 100% qcov, *median* 99.79% pident. Full range: qcov 96.54–100%, pident 99.70–99.95%. Fully consistent with the paper’s 100%

qcov / 93–98% pident on their broader 26-plasmid set; if anything our identity is tighter because our reference set was filtered strictly to IncX4/mcr-1/30–40 kb.

4.9 Per-claim what-worked / what-didn’t

- C1–C7: **fully reproduced** from independently pulled NCBI records; every quantitative claim (accession numbers, sizes, AMR gene identities) recovered at $\geq 99\%$ identity/coverage.
- C8: **partially reproduced** — 2 of 4 paper-claimed chromosomal VGs directly confirmed (*astA*, *lpfA*); the other 2 (*fyuA*, *hlyE*) are not in AMRFinderPlus’s VF panel. Same-pathway evidence supports *fyuA* via *ybtPQ*. This is a *tool-panel gap*, not a real disagreement, but the report should say so honestly.
- C9: **reproduced** on an independent 17-plasmid reference set; observed identity even tighter than the paper (99.70–99.95% vs paper’s 93–98%) because of stricter reference selection.
- C10: **not tested** — ~ 500 -genome cgMLST + proprietary Ridom SeqSphere+ is out of scope for a free-endpoint 15-minute rerun. Documented as an open question (Q2).
- C11: **not testable in silico** — wet-lab claim. Documented as open question (Q1).

5 Critique of the Replication

Per Rick’s 2026-07-05 hard requirement: honest self-critique, not rubber-stamp. This replication has **real limitations** that the REPORT.md wording downplays.

1. **We validated the *deposited assembly*, not the paper’s raw-read $\rightarrow$ assembly workflow.** All AMR / VF / MLST / serotype / plasmid-typing evidence in the rerun is derived from the CP134085–CP134090 records already sitting in NCBI. If those records are wrong (e.g. assembly errors, mis-scaffolding of the 190 kb hybrid plasmid where two duplicated cassettes could create scaffolding ambiguity, mis-polished Nanopore homopolymer indels), we would silently inherit the same errors. The paper’s actual pipeline (Illumina + ONT reads $\rightarrow$ Unicycler 0.5.0 + Nanopolish 0.14.0) was *not* independently re-run from SRA. This is a real evidence-strength cap, not a formality — “the deposited FASTA agrees with itself” is a weaker claim than “independent re-assembly from raw reads agrees with the deposited FASTA”.
2. **Tool panels are not comprehensive, and negative results were treated as strong evidence when they are not.** AMRFinderPlus 4.2.7’s *VF panel* does not include *fyuA*, *hlyE*, or *hlyF* (all named by the paper), and does not annotate colicins, bacteriocins, mobilization/transfer functions, or general ORFs on Col156 / p0111 plasmids. Our confirmation of C7 (“101 kb + 5 kb plasmids carry no AMR/VG”) is therefore a confirmation *only against the AMRFinderPlus curated panel* — not a confirmation that these two plasmids are biologically inert. The REPORT.md phrases this as “AMR-free status confirmed” which is over-strong. This is exactly the “tolerance hand-waving” Rick flagged in the brief. See Q5 for the follow-up.
3. **LLM-judge coverage of 85% is arithmetic, not epistemic.** The judge counted 8 of the tested claims (C1–C9) as covered and said 100% agreement on those. But C10 and C11 — the two claims the paper’s *Discussion* spends the most words on (the epidemiological cgMLST tree and the anomalous non-transferability of the mcr-1 plasmid) — are the ones we *didn’t*

test. So the “85% coverage / 100% agreement” number is systematically biased toward the easily-testable claims. The claims we skipped are the most interesting and most contested ones in the paper.

4. **Reference set for the backbone-conservation BLAST was hand-curated, and that biases the identity distribution upward.** We stripped the 20 research hits to 17 real IncX4/mcr-1/30–40 kb plasmids (dropped 3 near-duplicates); the paper’s set of 26 plasmids likely includes broader IncX4 diversity (some without mcr-1, some outside the 30–40 kb window). Our 99.70–99.95% identity is therefore *expected* to be tighter than the paper’s 93–98% because our filter is stricter. We should not read our higher-identity number as “even stronger evidence” — it’s the same evidence on a narrower reference set. The rerun *does* corroborate the qualitative claim (“highly conserved backbone”), but not more strongly than the paper.
5. **The 100-bp chromosome length disagreement (paper 4,966,963 vs NCBI 4,967,063) was written off as “likely paper typo”.** That is a plausible guess but we did not check the paper’s supplementary materials, did not compare the paper’s assembly version tag to the current NCBI .1 record (a re-polishing or re-submission could have added 100 bp), and did not read the NCBI version history. “Typo” is the working assumption; the actual root cause was not investigated.
6. **We did not check the deposited FASTA MD5 sums against any secondary NCBI mirror or the paper’s supplementary FASTA (if any was submitted).** If the NCBI records have been silently updated post-publication (which does happen for polished Nanopore assemblies) our “exact size match” on 5 of 6 replicons could be a match to a *revised* record, not the record as it stood when the paper was written.
7. **The LLM judge was single-model + single-shot after fallback.** Argo Opus 4.7 flapped with 502s, we swapped to Argo GPT-5.2 and accepted the first response. No cross-model agreement check, no re-run with a different prompt framing. For a REPLICATED verdict on a comparatively straightforward genomic replication this is defensible, but the verdict has $1\times$ redundancy, not the $2\text{--}3\times$ we would want for a marginal call.
8. **No wet-lab or expression evidence.** Every claim we tested is sequence-based. The paper’s most *clinically* loaded claims — colistin MIC 8 $\mu\text{g/mL}$ (Q4), non-transferability under conjugation (Q1) — cannot be touched by this pipeline.

Bottom line of the critique: the *REPLICATED* verdict is defensible for the sequence-based claims we tested (C1–C9), but the report should be honest that (a) we tested against curated tool panels not against biology, (b) the two most epidemiologically-loaded claims (C10, C11) were not touched, and (c) we validated the deposited assembly not the paper’s assembly workflow.

6 Verdict

REPLICATED. Every sequence-testable claim (C1–C9) independently reproduces on a live rerun of the paper’s genomic pipeline against independently-pulled NCBI records. C10 (cgMLST tree) and C11 (non-transferability) were declared out of scope. LLM judge `argo:gpt-5.2`: `{verdict: REPLICATED, coverage_pct: 85, agreement_pct: 100, one_line: “NCBI assemblies confirm Ec45-2020 ST162 O55:H10; mcr-1.1 only on IncX4 33.5kb plasmid; other plasmids/AMR largely match.”, confidence: high}`.

The critique in §5 does not change the verdict but does bound its interpretation: this is “the deposited assembly re-agrees with itself under an independent in-silico pipeline”. It is *not* “the paper’s raw reads independently re-assemble and re-confirm the assembly”; that stronger claim would require pulling the SRA reads and re-running Unicycler + Nanopolish, which is out of scope but achievable in a second wave.

7 Open Questions

Five open questions grounded in re-reading the paper (see `work/paper_text.md`, extracted from PMC10604428) plus what this replication surfaced. Full JSON in `report/open_questions.json`.

Q1. Molecular basis of non-transferability. Why is pEc45-2020-33kb non-transferable under standard conjugation when the same IncX4/mcr-1 backbone is transmissible in the near-identical Chinese/European plasmids (>99.7% weighted identity) it shares $\geq 96.5\%$ coverage with? Is it (a) oriT loss/mutation this replication has not localized, (b) trans-inhibition by the co-resident 254 kb IncH + 190 kb IncF plasmids (Szmolka’s “conjugation competition” speculation), or (c) strain-level T4SS regulatory silencing? Basis: paper’s own Discussion explicitly flags this as unresolved (“the oriT was not detectable ... warrants further investigation”); our BLAST rerun confirms the T4SS locus is essentially intact so the block is not gross sequence loss.

Q2. Robustness of the cgMLST cluster claim. Is the “Hungarian duck ST162 clusters with Chinese human ST162 not Chinese duck ST162” finding robust to (a) the 238-genome subsample, (b) the K-12 MG1655 reference choice, and (c) the proprietary Ridom SeqSphere+ v9.08 defaults? Basis: C10 is out of scope for our rerun; the paper shows a single tree with no bootstrap, no independent scheme, and no time-calibration — yet its “human $\rightarrow$ duck” direction claim depends on this tree.

Q3. Mechanism + phenotype of the duplicated blaTEM-135–sul2–tet cassette. What is the mechanism (transposition, homologous recombination, IS-mediated capture) and evolutionary timing (pre-existing vs recent independent acquisition) of the duplication of this cassette across CP134087 and CP134088, and does it raise effective AMR gene copy number / MIC? Basis: paper reports the duplication as bare fact; no flanking-IS analysis, no copy-number qPCR, no MIC-vs-single-copy comparison. AMRFinderPlus alone cannot distinguish recent duplication from independent acquisition.

Q4. Modest colistin MIC (8 $\mu\text{g}/\text{mL}$) vs typical mcr-1. Why is the colistin MIC exactly at the EUCAST resistance breakpoint (8 $\mu\text{g}/\text{mL}$) rather than the more typical mcr-1 range of 4–16 with mode ~ 16 ? (a) Low mcr-1 expression from the IncX4 promoter in this duck strain background, (b) compensating chromosomal mgrB/phoP/pmrAB polymorphism, or (c) fitness cost of the two large co-resident plasmids? Basis: paper reports MIC 8 without context or chromosomal mgrB/pmrAB screening; our AMRFinderPlus rerun does not comprehensively cover the E. coli colistin core-mutation panel. The modest MIC materially affects the One-Health interpretation.

Q5. Hidden gene content of the “AMR-free” 101 kb + 5 kb plasmids. Both the paper and this replication dismissed CP134086 and CP134090 as “no AMR/VG determinants” — but neither actually annotated their ORFs. Col156 plasmids commonly carry colicins; p0111 plasmids are implicated in APEC virulence in the literature. Basis: our AMRFinderPlus rerun only calls

curated AMR/VF panels. “No hits from AMRFinderPlus” $\neq$ “no biologically meaningful genes”. This is exactly the kind of “data not shown” gap Rick flagged.

Concrete next steps for each Q are enumerated in `report/open_questions.json`.

8 Companion Artifacts

- `report/REPORT.md` — original markdown report (kept per standard).
- `report/open_questions.json` — machine-readable version of §7.
- `report/workflow.md` — workflow narrative + tool inventory + effort estimate.
- `report/artifacts_summary.md` — inventory + traces.
- `report/failure_analysis.md` — honest failure/gap analysis.
- `report/evidence/` — `amrfinder_full.tsv`, `mlst_warwick.tsv`, `mlst_pasteur.tsv`, `plasmidfinder.tsv`, `serotype.tsv`, `blast_incx4_v1.tsv`, `blast_incx4_v2.tsv`, `blast_incx4_v2_summary.json`, `llm_judge_verdict.json`, `llm_judge_model.txt`, `amrfinder.log`.
- `work/` — `paper_pmc.xml`, `paper_text.md`, `pmc.link.json`, `pubmed_meta.json`, `mcr1_ref.faa`, `blast_summary.py`, `llm_judge_prompt.txt`, `genomes/`.
- `paper.pdf` — backfilled from MDPI open-access (CC BY 4.0) during 2026-07-05 8-artifact pass.
- `extraction/marker.md` — pdftotext fallback extraction; central Marker corpus copy pending sha256 resolution.
- `extraction/nougat.mmd` — pending central Nougat parse (GPU-only; stub with sha256 header written for later corpus sweep).